

Takeaways & Preview

- Dynamic OT = geodesic view of transport, links geometry, PDEs, and ML.
- Benamou–Brenier = kinetic energy minimization
- Continuity equation ensures mass conservation
- McCann interpolation = explicit geodesic formula
- Next lecture:
 - Gradient Flows
 - Entropic/stochastic variants (Schrödinger bridges)

Interactive Demo: Bunny Interpolation



tinyurl.com/cs2840-lecture8

1. Implement displacement interpolation:

```
def displacement_interpolation(t, XA, T):  
    """ Displacement interpolation at time t using  
        barycentric map T on source X_src.  
    Args:  
        t: float in [0,1]  
        XA: (N,3) source point cloud  
        T: (N,3) barycentric map from source to target  
    Returns: (N,3) interpolated point cloud at time t  
  
    """  
    # TODO: return the interpolated cloud at time t  
    # using X_src and T  
    # Fill in here  
    Xt = ...  
    return Xt
```

Distribution:

$$\rho_t = [(1 - t)\text{Id} + tT]_{\#}\alpha$$

Particle trajectories:

$$x_t = (1 - t)x + tT(x)$$

2. Choose ε values for Sinkhorn.

3. (Optional) Download mesh files for other shapes, try them out.